M201103W- Signals and Systems

ssch7_2 –z- transform of elementary signals

14:10 pm Wednesday @ C107, 1st Teaching Building

14:10 pm Friday @ A107, 1st Teaching Building

Hang Zhou

hangzhou@bjtu.edu.cn

z- transform of elementary signals

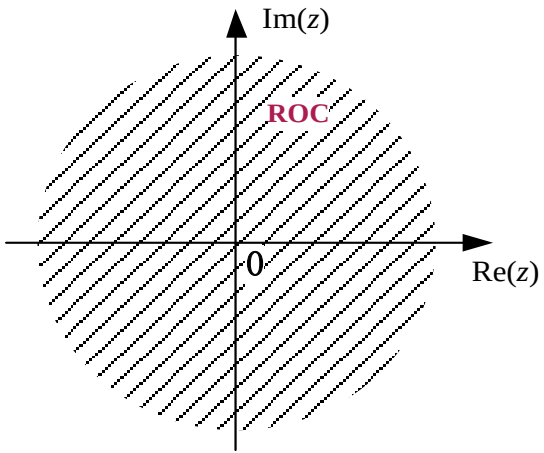
- Unit impulse signal
- Unit step signal
- Exponential signals
- Imaginary exponential signals
- Sinusoidal signals

Unilateral z- transform of elementary signals

1. Unilateral z-transform of $\delta[k]$

$$\mathscr{Z}\{x[k]\} = \sum_{k=0}^{+\infty} x[k]\, z^{-k} = \sum_{k=0}^{+\infty} \delta[k]\, z^{-k} = 1$$

$$|z| \geq 0$$



$$\delta[k] \xleftrightarrow{\mathscr{Z}} 1, \quad |z| \geq 0$$

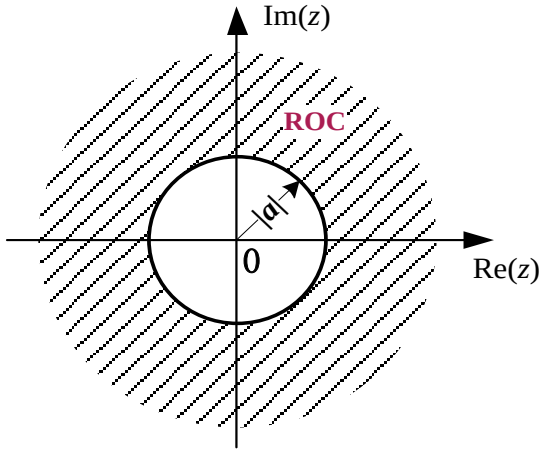
Unilateral z- transform of elementary signals

2. Unilateral z-transform of $a^k u[k]$

$$\mathscr{Z}\left\{a^k u[k]\right\} = \sum_{k=0}^{+\infty} a^k u[k]\, z^{-k}$$

$$= \sum_{k=0}^{+\infty} (az^{-1})^k = \frac{1}{1-az^{-1}}, \qquad |z| > |a|$$

$$a^k u[k] \xleftrightarrow{\mathscr{Z}} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z| > |a|$$

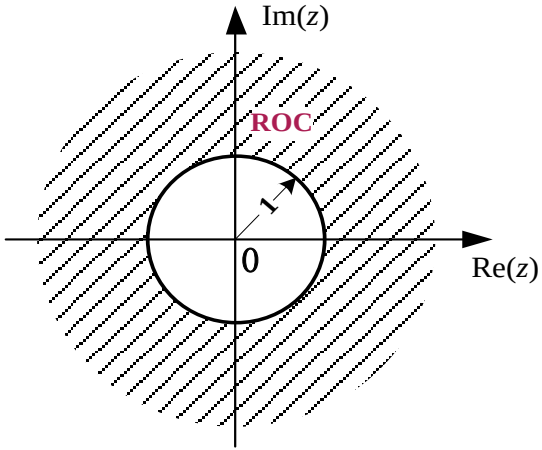


Unilateral z- transform of elementary signals

3. Unilateral z-transform of $u[k]$

Because $$a^k u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \qquad |z| > |a|$$

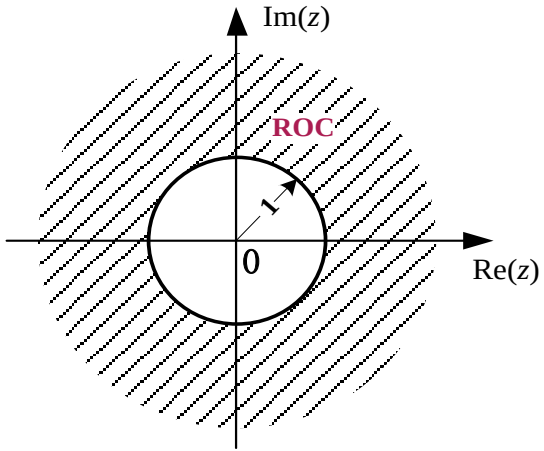
Let α=1 $$u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-z^{-1}} = \frac{z}{z-1}, \qquad |z| > 1$$



Unilateral z- transform of elementary signals

4. Unilateral z-transform of $e^{j\Omega_0 k}u[k]$

Because $$a^k u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-az^{-1}} = \frac{z}{z-a}, \quad |z| > |a|$$



Let $\alpha = e^{j\Omega_0}$

$$e^{j\Omega_0 k}u[k] \xleftrightarrow{\mathcal{Z}} \frac{1}{1-e^{j\Omega_0}z^{-1}}, \quad |z| > 1$$

Unilateral z- transform of elementary signals

5. Unilateral z-transform of $\cos(\Omega_0 k)\, u[k]$ and $\sin(\Omega_0 k)\, u[k]$

$$\mathscr{Z}\left\{\cos(\Omega_0 k)u[k]\right\} = \mathscr{Z}\left\{e^{j\Omega_0 k}u[k]\right\}/2 + \mathscr{Z}\left\{e^{-j\Omega_0 k}u[k]\right\}/2$$

Euler's relation

$$= \frac{1}{2}\frac{1}{1-e^{j\Omega_0}z^{-1}} + \frac{1}{2}\frac{1}{1-e^{-j\Omega_0}z^{-1}}$$

$$= \frac{1-z^{-1}\cos(\Omega_0)}{1-2z^{-1}\cos(\Omega_0)+z^{-2}}$$

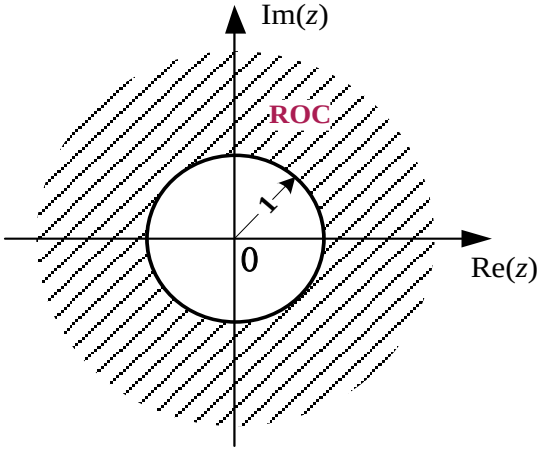
$$= \frac{z^2-\cos(\Omega_0)z}{z^2-2\cos(\Omega_0)z+1}, \qquad |z|>1$$

Unilateral z- transform of elementary signals

5. Unilateral z-transform of $\cos(\Omega_0 k)\, u[k]$ and $\sin(\Omega_0 k)\, u[k]$

$$\cos(\Omega_0 k)u[k] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{1-\cos(\Omega_0)z^{-1}}{1-2\cos(\Omega_0)z^{-1}+z^{-2}}, \quad |z|>1$$

$$= \frac{z^2-\cos(\Omega_0)z}{z^2-2\cos(\Omega_0)z+1}, \quad |z|>1$$



$$\sin(\Omega_0 k)u[k] \overset{\mathcal{Z}}{\longleftrightarrow} \frac{\sin(\Omega_0)z^{-1}}{1-2\cos(\Omega_0)z^{-1}+z^{-2}}, \quad |z|>1$$

$$= \frac{\sin(\Omega_0)z}{z^2-2\cos(\Omega_0)z+1}, \quad |z|>1$$